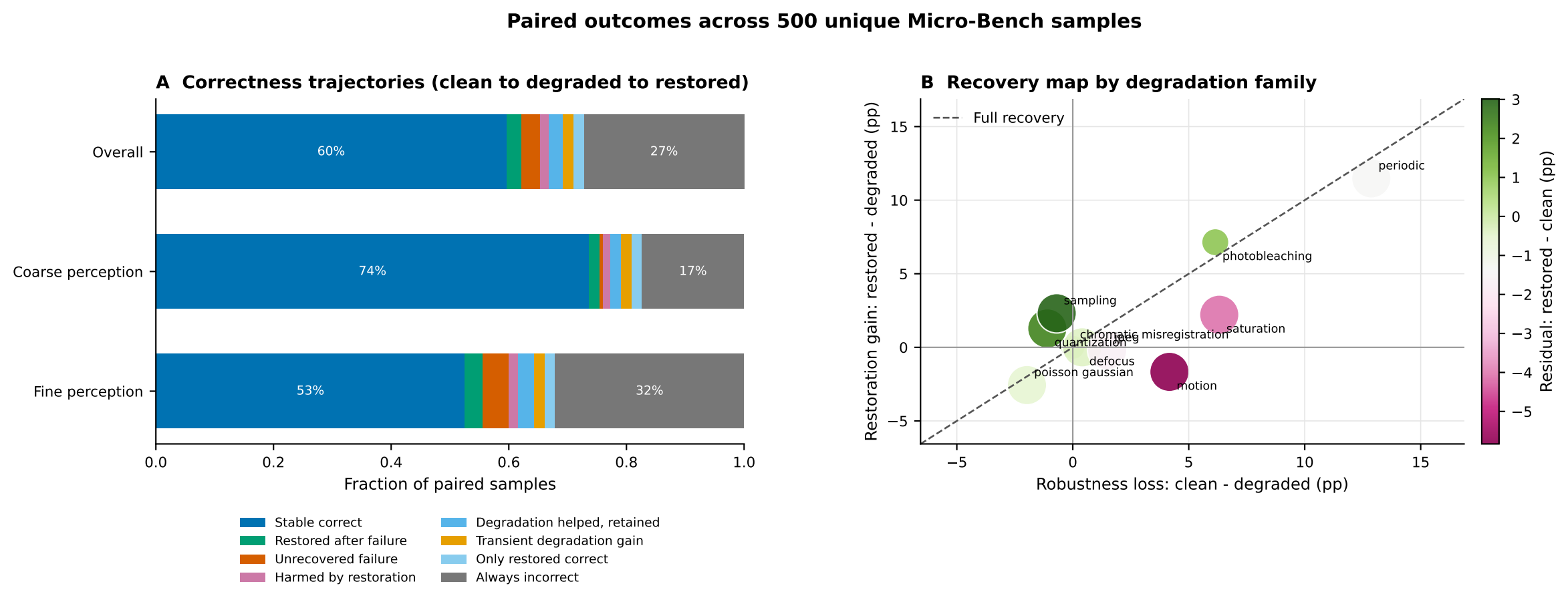
Paired outcomes across 500 unique Micro-Bench samples
A Correctness trajectories (clean to degraded to restored)
Overall
Coarse perception
Fine perception
60%
27%
74%
17%
53%
32%
0.0
0.2
0.4
0.6
0.8
1.0
Fraction of paired samples
Stable correct
Restored after failure
Unrecovered failure
Harmed by restoration
Degradation helped, retained
Transient degradation gain
Only restored correct
Always incorrect
B Recovery map by degradation family
Full recovery
Restoration gain: restored - degraded (pp)
Robustness loss: clean - degraded (pp)
Residual: restored - clean (pp)
periodic
photobleaching
sampling
saturation
chromatic misregistration
defocus
poisson gaussian
motion